Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Agentic AI systems engineer who builds evals alongside agents -- AgentProbe (open-source eval infrastructure: 5-metric engine, pass@k harness, CI regression gate, model routing) and LeaseI (8-agent LangGraph StateGraph for regulated compliance decisions with HITL, confidence scoring, and deterministic audit trail) -- with 8 years of production SWE depth in distributed systems, RabbitMQ idempotency, FastAPI APIs, and observability. Currently building ML models and data pipelines at Blue Horse Digital. Core differentiator: I don't just ship agents; I build the measurement harness that tells you when they're working and when they're not. AgentProbe: github.com/venkatalakshmiparimala/AgentProbe. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Agentic AI Systems + LangGraph Orchestration | Eval Infrastructure (Offline + Online + A/B) | LLM Reliability + Guardrails + HITL | Model Routing + Cost Controls | RAG Pipelines + Structured Outputs | Safety + Auditability + Policy Compliance | Distributed Systems + RabbitMQ Idempotency | Production AI + Observability + Tracing

WORK EXPERIENCE

Blue Horse Digital

Sep 2026 – Present

AI/ML Engineer (Volunteer) · Virginia, USA

- Developing ML models and data analysis pipelines for software applications and business intelligence projects -- from data collection and cleaning through feature engineering, model training, and evaluation.
- Integrating data systems and AI components with back-end and front-end teams; building pipelines that surface insights from raw business data into application-facing APIs and reporting layers.

Magizhs Tech Solution

May 2025 – Aug 2025

AI/ML Engineer Intern

- Built an agentic mission-planning system orchestrated as a LangGraph stateful graph (parse → plan → retrieve → validate → conditional routing) delivering a risk-aware compliance strategy: compliant missions route to export, non-compliant ones route to rejection with grounded reasons and cited source regulation.
- Implemented a RAG layer (retrieval over FAA Part 107 regulatory corpus) with function calling, structured outputs, and citation grounding -- every compliance check functions as a hypothesis test grounded in source regulation. Validation guardrails enforced altitude limits, battery budgets, and geometric no-fly-zone checks before export.

Entrata (PropTech / Real Estate SaaS)

Sep 2018 – Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Built and maintained RabbitMQ-based async processing system** for gross-rent updates across thousands of tenant records -- designed retry logic and idempotency guarantees that eliminated double-processing failures; owned through production incidents and annual rent-schedule changes.
- **Owned HUD income-limit import pipeline for 7 years** -- few thousand federal source rows expanded into 180,000 domain records powering eligibility, rent, HAP, and tax-credit calculations for 100K+ housing units at 99.5% uptime. Zero compliance violations across the ownership period.
- **Built 15-endpoint REST APIs** and optimized 200+ SQL queries (30% performance gain). Led microservices redesign using Docker -- 50% scalability improvement, 40% deployment time reduction. Observability via structured logging and performance engineering across all pipelines.
- **Led AWS S3 document migration** -- 100% PHI compliance with Libsodium-encrypted pipelines; zero security incidents. Mentored 5+ engineers; 300+ peer code reviews; debugging tools that reduced bug count below 10 for 6+ consecutive months.

George Mason University

Aug 2024 – Aug 2025

Research Associate & Teaching Assistant

- Built prediction models ($R^2 = 0.977$) and deployed inference API on Quart (async Python); Isolation Forest for automated anomaly detection across 3 pipelines. TA for 600+ students: Python programming, OOP, automation scripting, SQL database design.

Entrata India, Pune

Jul 2017 – Sep 2018

Associate Software Engineer

- Built compliance modules (Python, Java, PostgreSQL); led AWS S3 cloud migration; redesigned document storage architecture for hundreds of thousands of records with full validation.

PROJECTS

AgentProbe: Open-Source AI Evaluation Toolkit

Python · FastAPI · LangGraph · CI/CD

Built fully local, zero-cost AI agent evaluation infrastructure: 15-endpoint FastAPI backend, 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression harness with golden datasets, model routing across OpenAI/Anthropic/Groq/Ollama, and offline + online evaluation loops -- used to evaluate Leasel across 1,240+ compliance test cases. github.com/venkatalakshmiparimala/AgentProbe

Leasel: Agentic AI System for Regulated Decisions

LangGraph 8-agent · FastAPI · OCR · SQLite

8-agent LangGraph StateGraph for compliance investigation with confidence-based HITL. Core rule enforced in code: LLM extracts and reasons; human or deterministic logic makes every final regulated call. Evaluated via AgentProbe: 93.6% precision, 87.9% F1, pass@3 88.3% across 1,240 lease PDFs. github.com/venkatalakshmiparimala/Lease_complaince_rag

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · Docker · BERT

~60% reduction in manual clinical retrieval time -- NLP platform translating natural-language queries into FHIR API calls using LLMs, BERT, and embeddings; HIPAA-compliant RBAC with data isolation across structured medical records. github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur — 2017

SKILLS

AI & Agents: LangChain, LangGraph, Agentic AI, Multi-Agent Orchestration, Tool Calling, Function Calling, Structured Outputs, RAG Pipelines, Hybrid Search, Semantic Search, HITL, LLM Evaluation, LLM-as-Judge, Golden Datasets, Offline Evaluation, Online Evaluation, A/B Testing, Model Routing, Cost Controls, Guardrails, Hallucination Detection, Confidence Scoring, Auditability, Policy Compliance, Agent Observability, Trace-Level Observability, Prompt Engineering, OpenAI API, Anthropic API, Groq, Ollama, ChromaDB, Qdrant, MCP

Languages: Python, PHP, JavaScript, TypeScript, Java, Scala, SQL, PL/SQL, R, Bash

Backend & Systems: FastAPI, REST APIs, Spring Boot, RabbitMQ, Redis, Microservices, Distributed Systems, Idempotency, Async Processing, Multi-Tenant Architecture, Docker, Kubernetes, OpenShift, CI/CD (GitHub Actions)

Cloud & Observability: AWS (S3, EC2, RDS, Glue), Azure (ADF, ADLS, Databricks), Prometheus, Grafana, OpenTelemetry, Apache Spark, PySpark

Databases: PostgreSQL, MySQL, Oracle, Snowflake, Teradata, NoSQL, Solr

ML: PyTorch, TensorFlow, Scikit-learn, HuggingFace, XGBoost, Isolation Forest, Pandas, NumPy, spaCy, BERT